AdaptVR: Leveraging Reinforcement Learning and LLMs for Personalized Cybersickness Mitigation and Reasoning in VR

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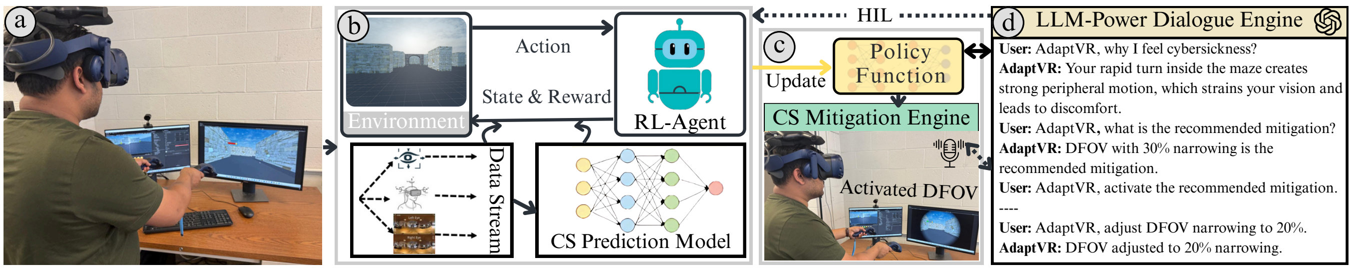

Figure 1: An overview of the AdaptVR framework. (a) A participant immersed in a VR maze simulation. (b) An RL agent observes the state from the environment/data stream, with rewards shaped by a cybersickness (CS) detection model. (c) The policy activates DFOV from the cybersickness mitigation engine. (d) The user interacts with an LLM-powered dialogue engine to adjust the DFOV aperture size, while the logged data is subsequently used for HIL fine-tuning.

Abstract

Cybersickness remains a persistent barrier to the widespread adoption of virtual reality (VR), often undermining user comfort and immersion. Existing cybersickness mitigation techniques are mostly static and may fail to adapt to diverse user needs and dynamic VR contexts. For instance, aperture size in the dynamic field of view (DFOV) and blur intensity in the dynamic Gaussian blur (DGB) may need to vary across users and applications based on real-time cybersickness severity. To address this, we introduce *AdaptVR*, a reinforcement learning (RL)–based adaptive cybersickness mitigation framework that predicts, explains, and mitigates cybersickness. Unlike static methods, our RL agent optimizes long-term user comfort by adaptively selecting and adjusting mitigation strategies through continuous interaction with VR environments. We train the RL agent using Proximal Policy Optimization (PPO) in domain-randomized simulations and deploy the learned policy on a consumer-grade VR headset (HTC Vive Pro). At runtime, AdaptVR continuously processes real-time user signals (e.g., eye-tracking, head motion, scene context) to detect cybersickness severity and appropriate mitigation (e.g., DFOV and DGB) with exact intensity. We also design a large language model (LLM)–powered interactive dialogue engine that enables users to engage in natural, voice-based conversations to understand detection outcomes, explore mitigation choices, and provide feedback. This feedback is incorporated into a human-in-the-loop training process, allowing the RL agent to fine-tune its policy for personalized cybersickness detection and mitigation. We evaluate AdaptVR through a user study in a VR maze simulation. Results show that our RL-driven adaptive mitigation significantly reduces cybersickness with minimal impact on immersion. Moreover, 90% of participants reported that the framework is intuitive, effective, and improved their understanding of the mitigation process through the dialogue engine. The open source implementation of the AdaptVR framework can be found at https://anonymous.4open.science/r/AdaptVR-D676.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; • **Computing methodologies** → *Reinforcement learning*; Natural language processing.

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1 Introduction

Virtual Reality (VR) has gained tremendous popularity across diverse domains, including healthcare [34], national defense [4], gaming [58], and beyond [48]. Despite its rapid growth, cybersickness remains a major obstacle to the broader acceptance and usability of VR systems [6, 25, 39]. Cybersickness manifests as symptoms such as nausea, disorientation, and visual fatigue, which significantly reduce user comfort and immersion, thereby limiting the scalability of VR applications. Prior studies have explored and applied different *static* cybersickness mitigation methods (e.g., manipulating the field of view (FOV), changing the depth of field (DOF) blur, etc.) by adding visual effects or elements to virtual environments to help alleviate the effects of cybersickness [5, 6, 12, 21, 25, 26, 30]. One well-known example is the dynamic FOV (DFOV) restrictor by Fernandes and Feiner [19], which narrows the outer FOV during

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rapid head rotations from wider ranges (e.g., 120°–155°) to narrower presets (e.g., 68°–120°) and then restores it afterward. Similarly, the dynamic Gaussian blur (DGB) technique applies different blur intensities during rapid head or camera movements by varying the Gaussian kernel's standard deviation σ (e.g., σ = 5–8 for mild motion and σ = 10–15 for fast motion) and restoring full clarity once the motion stops [5, 12]. However, such techniques are either manually applied, requiring human intervention, or limited to *fixed/static* presets, e.g., aperture size and blur intensity in the dynamic FOV, and the dynamic Gaussian blur remains predetermined. Hence, such static mitigation methods may disrupt immersion and reduce their effectiveness across different users since cybersickness is highly subjective [46, 61, 62].

Recently, Kundu et al. [37, 39] proposed an explainable AI (XAI)-guided mitigation framework that maps important features to predefined mitigation techniques. While their work aims to automatically trigger cybersickness mitigation in a trustworthy manner and leverages large language models (LLMs) for user interaction with the framework, it has a few limitations. For instance, the framework is constrained by rule-based logic and lacks the ability to adaptively adjust mitigation strength (e.g., how much to narrow DFOV or how strongly to apply dynamic Gaussian blur (DGB)). In addition, it does not personalize mitigations based on user feedback or individual tolerances, limiting its applicability across diverse users and contexts. For example, consider a user navigating a VR maze where the goal is to escape while collecting coins scattered throughout the environment. When rapid movement or head turns are detected, a rule-based mitigation system may apply DFOV using a fixed, narrow preset to reduce cybersickness. While this can reduce discomfort, the uniform preset also blocks peripheral vision, making it harder to notice nearby coins and disrupting gameplay. Although stronger mitigation may be appropriate during fast-paced motion, the same adjustment can feel excessive in slower exploration, and its impact further varies as individuals differ widely in how much visual alteration they can tolerate [58]. Moreover, RelaxVR's interactive dialogue engine, built on pre-defined grammar and parser rules rather than pre-trained LLMs (e.g., GPT -40) [2], is unable to provide human-centric reasoning about the causes of cybersickness and the rationale behind mitigation strategies, and fails to capture the diverse and evolving needs of users in VR contexts. These limitations highlight the need for an *adaptive framework* that not only triggers mitigation automatically but also adjusts its intensity in real time, based on both the context and individual user preferences. Achieving such intelligent adaptation requires a learning-based approach that can move beyond fixed rules and continuously refine mitigation strategies through interaction.

To address this critical gap in state-of-the-art (SOTA) cybersickness mitigation, this paper proposes *AdaptVR*—a Reinforcement learning (RL)-based adaptive cybersickness mitigation framework with an interactive LLM dialogue engine, as shown in Figure 1. AdaptVR employs an RL agent that adaptively triggers appropriate mitigation with exact intensity, guided by a DL-based cybersickness detection model that shapes the agent's reward for effective mitigation. In addition, an LLM-integrated dialogue engine enables users to query decisions and provide feedback through voice commands, ensuring that mitigation remains not only effective but also aligned with individual comfort needs. RL offers a strong foundation for this goal and has been applied in diverse VR domains, such as education, training, and healthcare, to deliver adaptive, user-centered experiences [16, 20, 51]. With its capacity for sequential decision-making, real-time policy updates, and optimization through interaction-driven feedback, RL has proven highly effective for personalization in interactive systems [16, 51]. These properties make RL particularly well-suited for personalizing cybersickness mitigation, where the system must determine not only when to trigger a mitigation, but also how strongly it should be applied in line with individual user preferences. Our key contributions are summarized as follows.

- We first develop a DL–based Transformer model that classifies cybersickness severity into four categories: *none, low, medium, and high*. The detected severity is then used to shape the RL agent's reward function to train RL-based adaptive mitigation. We use the open source *VRWalking* dataset [57] to train the Transformer model. Our results show that the Transformer model classifies cybersickness with a high accuracy of 95%.
- Next, we develop and train an RL agent in a domain-randomized VR maze simulation (with diverse variations in layouts, motion dynamics, and visual flows) to perform adaptive cybersickness mitigation using streaming data (e.g., eye and head tracking), visual, and task context. The agent is trained using the Proximal Policy Optimization (PPO) algorithm [56], where the cybersickness detection model is used to shape the reward. Based on this reward, the RL agent learns to adaptively select the most appropriate mitigation technique along with accurate intensity from the cybersickness mitigation library, which includes *DFOV* with an adjustable aperture size and *DGB* with tunable blur intensity.
- We design an interactive dialogue engine for cybersickness reasoning using pre-trained LLMs that enables VR users to communicate with the RL agent using natural language. Users can ask queries about RL agent actions (e.g., *why am I feeling cybersick?*), inquire about the causes of their discomfort, or specify mitigation preferences (e.g., adjust DFOV narrowing to 20%).
- Finally, we fine-tune the RL agent using human-in-the-loop (HIL) adaptation to personalize mitigation based on individual comfort preferences. Feedback collected through the LLM-driven dialogue engine (e.g., user-reported discomfort or requests for adjustments) is used to guide the fine-tuning process. This enables the RL policy to progressively align with each user's unique tolerance profile, resulting in more effective and user-centered cybersickness mitigation.

We validate AdaptVR's effectiveness through a user study in a VR maze simulation. Our results show that AdaptVR significantly reduces cybersickness with minimal impact on user immersive experience (UIX). By adaptively selecting mitigation techniques and their intensities, our approach outperforms static and rule-based methods, with 90% of participants reporting superior comfort. *To the best of our knowledge, AdaptVR is the first framework to combine RL-based adaptive cybersickness mitigation with continuous intensity tuning and an LLM-powered dialogue engine for real-time user interaction and personalization in VR. To support reproducibility and provide a reference framework for future cybersickness research, we* *release AdaptVR as an open-source[1] implementation that includes the VR maze simulation, RL training pipeline, and LLM dialogue engine. Our goal is to enable researchers and developers to systematically evaluate and extend adaptive mitigation strategies, enhancing user comfort and accelerating VR adoption.*

2 Related Works

This section briefly discusses and reviews the current understanding of cybersickness and several key aspects pertinent to our work: cybersickness detection, cybersickness mitigation, RL and LLMs in VR. The details are as follows.

Cybersickness Detection: Cybersickness refers to user discomfort in virtual environments, with symptoms such as dizziness, eye strain, and nausea arising from visual–vestibular mismatch or postural instability. Traditional approaches to measuring cybersickness relied on pre- and post-exposure questionnaires, such as the Simulator Sickness Questionnaire (SSQ) [7, 12, 13, 15, 18, 19, 21, 24, 30, 47, 68]. Building on these methods, recent studies now focus on automatic cybersickness detection using ML/DL models that utilize multimodal sensor data (e.g. eye, video data, etc.) [17, 22, 26–28, 38–41, 54, 61]. For example, Kundu et al. [40, 41] developed explainable ML/DL frameworks for predicting cybersickness severity using VR simulation data. In contrast, author [28] employed Transformer-based models for cybersickness detection. Moreover, Tasnim et al. [61] investigated personalization cybersickness detection techniques across users, while Zhu et al. [70] introduced a consumer-grade detection framework based on a multimodal graph neural network. Kundu et al. [36, 38] further advanced cybersickness prediction in immersive virtual reality using pre-trained large foundation models, with the latter extending prediction to multiple concurrent VR tasks. However, most prior studies focus on cybersickness prediction and do not address how effective mitigation can be achieved in immersive VR environments.

Cybersickness Mitigation: Several visual techniques have been proposed to reduce cybersickness during immersive experiences [6, 8, 12, 25, 37, 39, 58], which can be broadly classified into static and dynamic approaches, along with consideration of whether human intervention (HI) is required to activate them, as shown in Table 1. Most existing works rely on static mitigation techniques, which apply fixed, non-varying mitigations. For instance, Fernandes et al [19] proposed a subtle DFOV adjustment technique in which the FOV adjusts automatically during rapid motion, effectively reducing sickness without noticeably diminishing presence. Similarly, Kelly et al. [30] demonstrated that restricting the FOV and employing snap turning can mitigate cybersickness during navigation. Furthermore, Calandra et al. [12] developed a standardized testbed that integrates multiple mitigation techniques (e.g., FOV adjustment and Gaussian blur, etc) for systematic evaluation across immersive VR scenarios. However, these methods are typically static, predefined, and often require HI (e.g., manual activation), which disrupts immersion in VR. To move beyond fixed, user-triggered approaches, a few studies have introduced dynamic mitigation strategies. For instance, Islam et al. [26] adjust mitigation techniques based on cybersickness severity. However, it still requires manual user input to activate certain responses and lacks full automation. On the other hand, Kundu et al. [37, 39] proposed an XAI-guided mitigation framework that dynamically maps important features to predefined mitigation techniques. However, it does not support adaptive tuning, as its parameters remain rule-based rather than continuously adjustment. Beyond visual methods, haptic approaches such as neck muscle vibration [60] and foot-based vibrotactile feedback [43] have shown significant cybersickness reduction, but these do not adapt to real-time cybersickness severity or individual user preferences. As summarized in Table 1, none of the existing systems provides adaptive mitigation, which involves continuous, real-time adjustment of mitigation parameters in response to user state. Instead, most remain static, require HI, or rely on hardcoded rules that fail to generalize across users or contexts. To address this, we propose AdaptVR, the first cybersickness mitigation framework that is dynamic and adaptive, operating entirely without HI.

RL and LLMs in VR: RL has been increasingly explored for adaptive, user-centered VR applications, such as motor rehabilitation [51] and neuroadaptive haptic feedback [20]. Prior studies also highlight RL's closed-loop adaptability and long-term personalization potential in healthcare and user modeling [16, 55]. However, none of these works apply RL specifically to cybersickness mitigation while preserving UIX via context-sensitive adjustment of visual parameters (e.g., FOV aperture, blur intensity). Beyond selecting cybersickness mitigation techniques with adaptive intensity adjustment, effective deployment in immersive systems also requires transparent, user-centered communication to maintain trust. To address this, recent works have explored integrating dialogue engines and LLMs into VR [9, 14, 23, 37–39, 42, 45, 49, 50, 59, 66]. For instance, authors in [23] introduced an NLP-based dialogue engine to monitor user cybersickness, but it lacked explicit mitigation techniques. Similarly, Kundu et al. [37, 39] incorporated LLM-based explanations into XAI-guided mitigation engine, but relied on predefined grammar and parser rules, limiting human-centric reasoning and adaptability. In contrast, AdaptVR integrates a pre-trained LLM-driven dialogue engine directly into the RL agent, enabling real-time reasoning about cybersickness severity, policy actions, and adaptive mitigation through natural-language interaction.

3 Proposed AdaptVR framework

In this section, we present the details of our proposed AdaptVR framework, illustrated in Figure 2. AdaptVR is implemented as a Unity project compatible with eye and head-tracking-enabled VR headsets (e.g., HTC VIVE Pro eye). We utilize the Tobii API [1] to stream real-time eye and head-tracking data, both for cybersickness severity detection to shape the RL agent's reward function and for the RL policy to trigger appropriate mitigation with accurate intensity. Furthermore, VR users can also interact with an LLM-guided interactive dialogue engine to understand RL policy's decisions, query cybersickness reasons, and set preferences for HIL personalization. It is important to note that RL agent training, HIL fine-tuning, and the interactive dialogue engine are executed in a cloud-based environment to ensure scalability and computational efficiency. The details are described in the following sections.

[1] https://anonymous.4open.science/r/AdaptVR-D676

Table 1: Comparison of prior work on cybersickness detection and mitigation with our proposed AdaptVR. Detection is either ML/DL-based or pre/post-questionnaire. Mitigation types include Static (S): fixed, non-varying techniques, Dynamic (D): rule or model-driven conditional adjustments and Adaptive (A): continuous, user-centric adjustments. Human intervention (HI): indicates whether user intervention is required to trigger mitigation.

Prior work	Detection	Mitigation			
		S	D	HI	A
Dissanayake et al. [17]	ML	×	×	×	×
Kundu et al. [40]	ML	×	×	×	×
Jeong et al. [28]	DL	×	×	×	×
Qi et al. [52]	DL	×	×	×	×
Zhu et al. [70]	DL	×	×	×	×
Kundu et al. [41]	ML/DL	×	×	×	×
Tasnim et al. [61]	ML/DL	×	×	×	×
Kundu et al. [38]	ML/DL	×	×	×	×
Croucher et al. [15]	Pre/Post	✓	×	✓	×
Nie et al. [47]	Pre/Post	✓	×	✓	×
Calandra et al. [12]	Pre/Post	✓	×	✓	×
Kelly et al. [30]	Pre/Post	✓	×	✓	×
Fernandes et al. [19]	Pre/Post	✓	×	✓	×
Ang et al. [7]	Pre/Post	✓	×	✓	×
Wienrich et al. [68]	Post	✓	×	✓	×
Hussain et al. [25]	Post	✓	×	✓	×
Groth et al. [21]	Post	✓	×	✓	×
Fantin et al. [18]	Post	✓	×	✓	×
Carnegie [13]	Post	✓	×	✓	×
Hadadi et al. [22]	ML	✓	×	✓	×
Chandra et al. [54]	ML	✓	×	✓	×
Islam et al. [26]	DL	×	✓	✓	×
Kundu et al. [37, 39]	ML/DL	×	✓	×	×
Our Proposed	ML/DL	×	✓	×	✓

3.1 Cybersickness Detection Model for RL Reward Shaping

For effective mitigation, it is crucial to detect cybersickness accurately, as we use these detection outcomes to shape the reward signals for the RL agent. Accurate detections allow the RL agent to correctly evaluate its mitigation actions over time. Thus, we first develop an accurate cybersickness detection model as a prerequisite for RL-based adaptive mitigation. We employ an SOTA DL-based Transformer model, which has been widely used in recent cybersickness detection research [28, 38, 39]. In our work, we use a Transformer model with positional encoding [65], a multi-head self-attention encoder, and an output layer to classify cybersickness severity (e.g., none, low, medium, high) following prior work [28, 32, 41]. We use categorical cross-entropy as the loss function for the cybersickness classification model. This classification provides a reward for RL-based mitigation methods, which we describe in detail in the following section 3.2. Note that the cybersickness detection model is trained and validated using 10-fold cross-validation. Afterwards, the trained model is exported to ONNX format and integrated within the RL-based mitigation method.

3.2 RL agent architecture and baseline training for adaptive cybersickness mitigation

The RL agent in our AdaptVR framework is designed to adaptively select mitigation strategies that minimize cybersickness while preserving visual comfort during immersive VR experiences. The agent continuously interacts with the environment by observing streaming data (e.g., eye and head tracking), visual flow, and task context, and then triggers mitigation actions based on the observed outcomes. We formalize this process as a Markov Decision Process (MDP), represented by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where each component is instantiated as follows:

State ***($s_t \in \mathcal{S}$):*** The state s_t encodes features relevant to cybersickness at timestep t, including locomotion signals (e.g., controller-driven angular velocity, linear acceleration), head-tracking signals (e.g., position, rotation, etc.), eye-tracking signals (e.g., pupil diameter, gaze direction, etc.), visual (e.g., scene dynamics), and task-level context (e.g., walking). To capture temporal dependencies, auxiliary variables (e.g., previous mitigation and its intensity) are also incorporated.

Action ***($a_t \in \mathcal{A}$):*** At each timestep t, the agent selects an action $a_t = (\mathrm{M}_t, I_t)$, where M_t specifies the mitigation type (e.g., DFOV or DGB), and I_t denotes the strength of the mitigation (e.g., DFOV aperture size or DGB intensity).

Transition dynamics ***(P):*** The transition probability function $P(s_{t+1} \mid s_t, a_t)$ defines how the environment evolves when an action is applied, mapping the current state and chosen action to the subsequent state. This captures how the user's physiological and perceptual state, such as motion, eye activity, and cybersickness severity, evolves over time under different mitigations.

Reward ***($r_t \in \mathcal{R}$):*** The reward leverages cybersickness severity detected by the Transformer-based classifier, as described in Section 3.1. Detected severity is then used to construct the reward, which encourages the agent to mitigate cybersickness. The RL reward function r_t at each time step t is defined as:

$$\begin{aligned} r_t &= (CS_{t-1} - CS_t) - \lambda_m I_t - \lambda_\Delta |I_t - I_{t-1}| \\ &-\lambda_{\text{switch}} \cdot \mathbf{1}[\mathrm{M}_t \neq \mathrm{M}_{t-1}] + \lambda_{\text{stab}} \cdot (1 - |I_t - I_{t-1}|) \end{aligned} \tag{1}$$

Here, CS_t denotes the continuous cybersickness severity at time t, obtained as the expected value over the Transformer's softmax output across the four severity levels. $I_t \in [0, 1]$ represents the normalized mitigation intensity, where both DFOV aperture fraction and DGB blur strength are mapped to $[0, 1]$. The selected mitigation technique is indicated by M_t. The weighting factors are defined as follows: λ_m penalizes excessive mitigation strength, λ_Δ penalizes abrupt changes in intensity (e.g., DFOV aperture size, DGB intensity), λ_{switch} penalizes switching between techniques, and λ_{stab} encourages smooth adjustments over time.

Discount factor ***(γ):*** The discount factor $\gamma \in [0, 1]$ sets how the agent balances short-term versus long-term effects of its actions. A small value prioritizes immediate cybersickness reduction, while a large value favors future comfort and stability.

The objective of the RL agent in our framework is to learn a policy $\pi_\theta(a_t \mid s_t)$ that selects a mitigation action a_t based on the current state s_t, interacts with the environment through transitions $\mathcal{P}$, receives rewards r_t for the Transformer model, and maximizes the expected cumulative return discounted by γ. To train the agent, we use PPO, a policy-gradient algorithm suited for high-dimensional

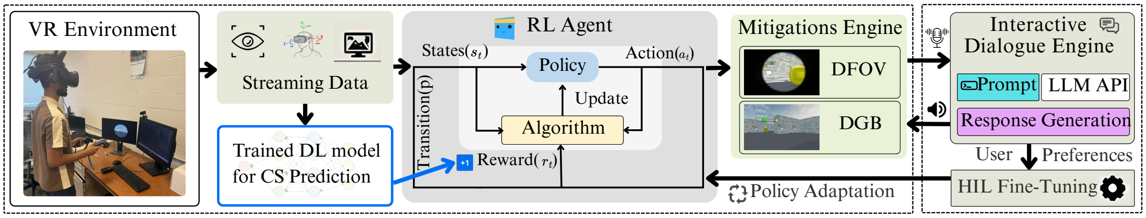

Figure 2: Overall architecture of AdaptVR for adaptive cybersickness detection, mitigation, and LLM-powered conversational interaction.

states and continuous or mixed actions [56]. PPO stabilizes training by constraining the size of policy updates while still encouraging sufficient exploration. Both the policy and the value function (which estimates expected future reward) are parameterized by neural networks [56], enabling the agent to generalize across diverse VR contexts. The effectiveness of PPO depends on several hyperparameters: the clipping ratio (ϵ), which bounds the magnitude of policy changes, the entropy coefficient (β), which promotes exploration and prevents premature convergence, and the learning rate (η), which controls the step size of gradient updates [53]. In our framework, these PPO components are applied to train the agent within a domain-randomized VR maze environment, which exposes the policy to diverse motion dynamics, visual flow, and task contexts. Training proceeds over mini-batches and multiple epochs using the PPO configurations as shown in Figure 5, while rewards are computed using Eq. 1. Note, in all experiments, we set $\lambda_m = 0.10$, $\lambda_\Delta = 0.05$, $\lambda_{\text{switch}} = 0.20$, and $\lambda_{\text{stab}} = 0.10$. The PPO settings and reward weights are selected via short pilot runs and remain stable under perturbations up to ±30%, with no major changes in convergence trends or final policy performance. We use early stopping to reduce overfitting, and validate the final policy in unseen randomized maze scenarios to assess generalization.

3.3 RL-Based Adaptive Mitigation Engine

During runtime, the RL policy observes the current state and selects the most suitable mitigation action along with its intensity to reduce cybersickness. This selected action is then passed to the mitigation engine, which in this study, is limited to two popular cybersickness mitigation techniques: DFOV and DGB. These techniques are commonly used in prior cybersickness mitigation research due to their proven effectiveness [3, 11, 19, 39, 44]. The DFOV applies a virtual filter to block peripheral vision, centered on the user's gaze [3], while DGB reduces discomfort by limiting optical flow [11]. It is important to note that, unlike prior studies using fixed presets under predefined triggers [19], our framework treats DFOV and DGB as fine-grained adaptive strategies that adjust continuously to context and individual tolerance. We implement DFOV using Unity's Tunneling Vignette Controller [63], where the Aperture Fraction (AF) parameter determines tunnel size. As cybersickness severity increases, the RL policy progressively reduces AF, narrowing the visible aperture, as shown in Figure 4(a). DGB is implemented using Unified Universal Blur [35], an open-source Unity package for efficient blurring within the Universal Render Pipeline (URP). Here, the RL policy regulates the intensity parameter to control blur strength, as shown in Figure 4(b). Although we only focus on these two techniques, other cybersickness mitigation with adjustable strength parameters (e.g., DOF, colorBlur, etc.) can also be integrated into

Based on the <**known information**> from the current VR environment and <**user preferences**>, please explain the <**flagged cybersickness**> that serves as the RL reward shaping function. Then provide the <**recommended mitigation**> and its <**recommended intensity**> from the RL agent for the user, considering a <**cybersickness severity level**> ranging from 0 to 3, where higher values indicate greater cybersickness severity. Ensure that the reasoning and the recommended mitigation strategy, including its intensity, align with user preferences.

Figure 3: A sample prompt template for LLM-based conversation generation in the AdaptVR framework.

our mitigation library. Each of these mitigations has its own advantages and disadvantages. Cybersickness severity can vary greatly between individuals, so a single fixed mitigation strategy may not suit everyone [58]. Our RL-based mitigation addresses this by selecting the most suitable action $a_t = (M_t, I_t)$ determined by both context and user preferences. The effectiveness of each choice is evaluated through the reward function 1. Through repeated interactions, the reward signal guides the policy toward context-dependent strategies, consistent with prior findings that the effectiveness of mitigation depends on the type of motion. For example, DFOV has been shown to reduce discomfort during rapid rotations [19], whereas DGB is more effective in attenuating optic-flow conflicts during forward motion [11].

3.4 Interactive Dialogue Engine

The interactive dialogue engine provides a natural language interface that makes the RL policy's mitigation actions and underlying cybersickness reasons explainable and interactable. Within this framework, the RL agent selects the appropriate mitigation technique with exact intensity, while the LLM supports preference articulation, explains the reasoning behind cybersickness and mitigation decisions, and collects user feedback for HIL fine-tuning. First, it takes inputs from the RL actions, predicted cybersickness severity level, and user state, along with user queries (e.g., *"why I am feeling cybersick?"*) via voice command. Then, the pre-trained LLM is used to generate responses to user queries. For this, we use a pre-trained GPT-4o mini LLM [2] with 8 billion parameters and apply the in-context learning (ICL) within the prompting paradigm [10, 67]. ICL enables the LLM to adapt dynamically to new tasks, such as explaining cybersickness severity or justifying RL policy decisions. By including a few task-specific examples within the prompt, the model can effectively understand the immediate context and generate human-centric and context-aware conversations. The dialogue engine allows users to naturally interact with the system through questions and feedback. For example, when a user ask questions (e.g., *"why I am feeling cybersick?"*), the engine analyzes the predicted severity level and the RL agent's recent actions, and the LLM provides clear reasoning for the detected discomfort. After receiving the reasoning, users request the RL-based recommended mitigation technique (e.g., *"What is the recommended mitigation technique?"*) and receive the adaptive mitigation technique with accurate intensity. In addition, users can also issue preferences (e.g., *"please set DGB intensity to 15%."*), which the engine interprets as constraints to update the runtime preference vector. These adjustments take effect immediately and are logged for offline fine-tuning, allowing the RL policy to gradually reflect long-term user preferences. Furthermore, prompt engineering techniques are used to refine user queries on cybersickness, guiding the LLM toward precise, context-aware conversations. We employ zero-shot prompting to extract features from interaction sequences and generate human-understandable explanations. The reasoning prompt template is shown in Figure 3. Note that for real-time explanation generation, we set the temperature to 0.2 and limited the output tokens to 400 for the GPT4o-mini. This approach ensures trustworthy and sufficient diversity in responses, making them suitable for effective decision-making for the users.

3.5 Personalized Cybersickness Mitigation via Human-in-the-Loop Training

In this section, we propose a personalized cybersickness mitigation approach that adapts to individual user preferences and tolerance levels. Although the baseline RL policy is already capable of adaptively triggering mitigations and adjusting their intensity, comfort thresholds can vary across users [58]. For example, some may prefer less DFOV narrowing or tolerate only minimal blur. To address this, we incorporate HIL fine-tuning, which leverages user feedback to refine the learned policy via a user-specific *preference vector* **z** [55] that encodes comfort limits (e.g., aperture size, intensity) and preferred trade-offs among mitigation techniques. This preference vector is utilized through four steps: (1) *Feedback logging*, where user-requested intensity overrides via the dialogue engine are stored in **z**; (2) *Reward weight update*, where after each session λ_m and λ_Δ (Eq. 1) are adjusted proportionally to logged preferences (e.g., a user who consistently reduces blur intensity causes λ_m to increase); (3) *Policy fine-tuning*, where a short PPO pass (10–20 episodes) updates the existing policy without resetting learned weights; and (4) *Soft-constraint enforcement*, where at runtime the bounds in **z** clip the agent's selected intensity to the user's tolerance range before rendering. For example, if a user prefers lower DGB intensity, the framework reduces blur strength slightly while preserving adaptivity, ensuring both comfort and responsiveness. This enables the agent to maintain mitigation effectiveness while continuously adapting to individual user tolerance.

4 Datasets, Experimental Setup & System Benchmarks

This section outlines the dataset and experimental setup used to validate the AdaptVR framework.

Dataset: This study uses the open-source VRWalking dataset [57], which includes eye-tracking, head-tracking, and physiological data from 39 participants (23 male, 16 females; mean age: 25.67, standard deviation (SD) = 7.22) from diverse ethnic backgrounds. Participants navigated a virtual maze by real walking for 15 minutes. The dataset provides eye-tracking features (e.g., Gaze Direction, Pupil Diameter, etc.), head-tracking features (e.g., Vector, Position, Rotation), and

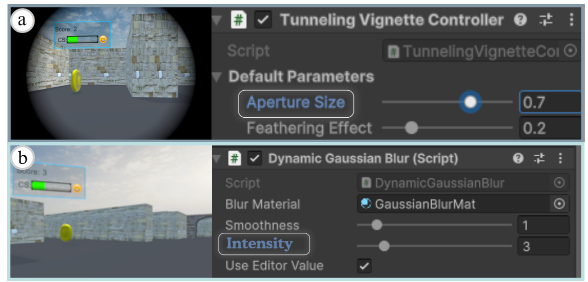

Figure 4: Visualization of mitigation techniques. (a) DGB with intensity controller, and (b) DFOV with aperture size controller.

physiological signals (e.g., HR, GSR), with ground-truth labels for six multitasking metrics: cybersickness, cognitive physical load, cognitive mental load, working memory, attention success rate, and attention reaction time. In this work, we specifically focus on cybersickness, as our framework requires reliable severity signals to shape the RL reward for cybersickness mitigation. Thus, following prior studies [40, 41], we categorized FMS scores (0–10) into four cybersickness severity levels: none, low, medium, and high. It is worth mentioning that for training the cybersickness detection model, we consider only integrated sensor data (e.g., eye and head tracking) rather than external physiological signals (e.g., HR), as these sensors require tethering or direct attachment to the user's body, which can introduce signal noise [27, 33].

Experimental Setup: To validate our proposed AdaptVR framework, we use a VR maze simulation similar to the one used for collecting the VRWalking dataset [57]. The RL agent is trained and evaluated using the Unity ML-Agents Toolkit [29], while cybersickness detection and reward shaping are implemented using PyTorch(2.3). Both the RL agent for adaptive mitigation and the DL-based cybersickness detection model for reward shaping are trained on a workstation with an Intel Core i9 CPU, 128 GB RAM, and an NVIDIA GeForce GTX GPU. To improve policy generalization across diverse VR environments, we applied domain randomization over maze topology (e.g., corridor width and junction density), wall-texture complexity, locomotion-speed ranges, and lighting conditions. The system output displayed on an HTC Vive Pro Eye HMD (2880 × 1600 per eye, 90 Hz) equipped with integrated eye tracking and head tracking

System Benchmarks: To evaluate the end-to-end latency of the AdaptVR framework, we measure the time from streaming data to rendering the RL-based recommended mitigation during immersion. To minimize cloud-processing delays, all backend services, including the LLM-powered dialogue engine, were deployed on an edge cloud server within the local VR network, ensuring rapid data exchange. During autonomous RL-based adaptive mitigation (without user intervention), the system is optimized for speed and achieves an average latency of ∼ 27 ms. In contrast, during user intervention, the voice-bas ed LLM dialogue system allows users to inquire about the reasons for cybersickness, receive explanations, and select RL-based preferred mitigation techniques. As a result, the interactive mode incurs a higher average latency of ∼ 271 ms due to query processing, explanation generation, and mitigation rendering. Still, it remains within acceptable limits for real-time VR interaction [39, 64].

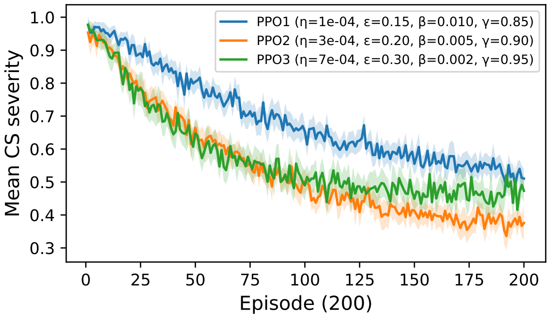

Figure 5: RL learning curves showing the mitigation of mean cybersickness (CS) severity across 200 training episodes.

5 Technical Evaluation Results

This section presents the technical evaluation of AdaptVR framework, including the cybersickness detection model for RL reward shaping, RL–based adaptive mitigation, and conversation-driven interaction for explanation and preference alignment.

Cybersickness Detection Model for RL Reward Shaping: Before integrating the Transformer model as a surrogate reward signal during RL agent training, we first report the development results with their statistics, such as training and inference time. The trained model has a size of approximately 360, 413 KB, with a total training time of about 725 minutes. For inference on the HTC Vive Pro Eye, the model requires 0.0021 seconds per frame to predict cybersickness severity from streaming eye and head-tracking data. The cybersickness classification using the Transformer model resulted in an overall accuracy of 95%, with precision values of 99%, 95%, 82%, and 100%; recall scores of 98%, 93%, 90%, and 96%; and F1-scores of 97%, 96%, 95%, and 96% for the none, low, medium, and high cybersickness classes, respectively.

RL Agent Performance: We evaluate the trained RL policy π_θ across $N = 200$ simulation episodes. On average, the policy requires only 2.0 ms to compute an action (e.g., DFOV or DGB) per frame. Given that a 90 Hz VR display refreshes every 11.1 ms, the RL agent consumes less than 20% of the available frame time. This confirms that the agent can operate comfortably within real-time constraints, while leaving sufficient headroom for rendering and other VR processes. We compare three representative PPO configurations, denoted PPO(1)–PPO(3), as shown in Figure 5, to analyze sensitivity to optimization parameters. Each of these PPO configuration corresponds to a different set of PPO hyperparameters (η, ϵ, β,γ), as introduced in Section 3.2. These configurations allow us to examine how hyperparameter choices influence convergence speed, training stability, and cybersickness mitigation. The selected values of these PPO configurations are drawn from ranges commonly recommended in prior PPO work [56] and widely used implementations such as Stable-Baselines3 [53]. In practice, we perform preliminary pilot sweeps within these ranges to ensure stable convergence in our VR environment. While all three configurations reduce cybersickness over time, their convergence rate and stability differ: PPO(3) outperforms PPO(1), and PPO(2) achieves the most balanced performance.

Conversation Generation and Understanding: We also demonstrate how well AdaptVR accurately generates conversations using the pre-trained LLMs and trained RL agent. These LLM-based human-centric conversations show the effectiveness of AdaptVR in understanding the cybersickness reasons through a dialogue engine. The overall conversation between the VR user with the cybersickness model and RL policy for understanding the reasoning behind the flagged cybersickness and how to choose effective mitigation techniques is illustrated with an example in Figure 6. Figure 6 shows that the VR user asks queries through a voice command (i.e., "Why I am feeling cybersick?") to know the reasoning behind flagged cybersickness, and AdaptVR provides the reasoning (i.e., *"You are moving through the maze, and the rapid optic flow of walls and textures in your periphery feels overwhelming."*) The user then asks for the recommended mitigation technique (i.e., "What is the recommended mitigation?"), and AdaptVR recommends DGB at 25% intensity, adaptively selected by the RL agent from the mitigation library. After feeling better, the user asks to stop the recommended mitigation, and AdaptVR deactivates the applied mitigation. Additionally, the user can request adjustments based on their preferences. It is worth mentioning that we demonstrate the user and dialogue engine's conversation on the VR simulation screen only for visualization purposes. However, in the experiment and user study, the user only sends queries to the LLM model through voice commands and listens to the generated responses to these queries, which provides better UIX without obstructing their VR immersion. *To see more conversations between the VR user and AdaptVR for understanding the reasoning behind cybersickness and how to choose effective mitigation techniques with intensity, please refer to the supplementary material.*

6 User Study Design on Cybersickness Mitigation and Evaluation

This section presents the detailed user study design and results of the user study evaluation to demonstrate the effectiveness of our proposed AdaptVR framework.

6.1 User Study Design on Cybersickness Mitigation

To demonstrate the effectiveness of the AdaptVR framework, we conducted a user study, and the details are described in the following subsections.

6.1.1 Participants, Apparatus and Virtual Environment. This user study involves 37 volunteer participants who received no reward (23 males and 14 females) aged from 22 to 38 years (mean age 26.8, SD 3.07). *It is worth mentioning that the participants' pool/sample size in our study is comparable to the prior cybersickness mitigation research [12, 19, 25, 37, 39].* The participants were Asian, Black, and Hispanic. Eleven of the thirty-seven participants reported having previously used VR equipment, whereas the rest reported never having done so. No individuals reported vestibular dysfunction or the use of anti-motion sickness medication. This study used a maze-based VR environment that allowed smooth, controller-based locomotion and visual exploration. Participants experienced the maze simulation in a standing condition to explore it. Participants could freely orient their view and controlled movement with hand-held controllers, and they were additionally tasked with collecting virtual coins placed throughout the maze. We rendered the VR simulations using the HTC Vive Pro Eye HMDs. The headset provided

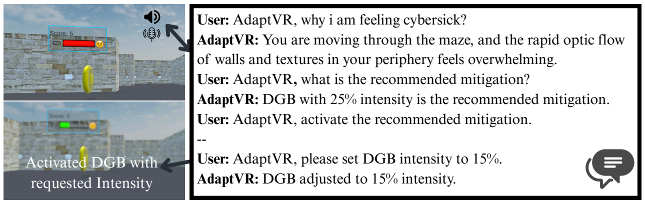

Figure 6: Conversation with AdapVR through an LLM-guided dialogue engine.

a wide field of view, spanning 110 degrees. Audio was delivered through the integrated Vive headphones.

6.1.2 Hypotheses. This research aims to identify and understand strategies that can effectively mitigate cybersickness. Therefore, the following three **hypotheses (H)** are formulated to form the basis of an experiment and evaluate the proposed AdaptVR cybersickness mitigation framework against traditional static or fixed mitigation techniques.

- H1: RL-based adaptive mitigation will reduce users' cybersickness symptoms more effectively compared to static mitigation.
- H2: Participants will have a better UIX in the virtual environment using adaptVR for cybersickness mitigation compared to other techniques.
- H3: Personalized mitigation through HIL adaptation will reduce cybersickness more effectively than both non-personalized and static techniques.

6.1.3 Study Task and Procedure. Our user study was approved by the Institutional Review Board (IRB) at the authors' university, the committee responsible for reviewing experiments involving human participants, ensuring that all ethical guidelines were followed, particularly with regard to participant safety and well-being. During the user study, the participants filled out the IRB-approved demographic information, such as age, gender, initial SSQ, and whether they had prior immersive VR experience. Afterward, they entered the VR simulation study. The details are described below.

Warm-up Phase: Before starting the experiment, we briefly introduced the procedures and informed participants about the requirements. We emphasized verbally reporting their discomfort on an FMS score of 0 to 10 whenever a pre-recorded voice prompt was played at each 30 seconds interval during the VR simulations. We chose to use an FMS scale from 0 to 10 to measure participants' cybersickness level in real-time and to be comparable to previous research in DL-based cybersickness detection [27, 39]. A rating of 0 means no significant change in comfort level compared to the resting baseline (no sickness), and 10 means significant discomfort compared to resting conditions (severe sickness). It is important to note that we used the FMS score instead of SSQ and VRSQ to measure cybersickness because SSQ and VRSQ are post-immersive questionnaires and do not provide real-time (during VR immersion) measurement of cybersickness. At the beginning of each trial, participants completed a brief practice session to familiarize themselves with different locomotion techniques.

Experiment Phase: After completing the trial session, the VR maze simulation were played and lasted 10 minutes long unless the participants decided to quit earlier. It is important to note that the eye and head positions were calibrated before starting the simulation. Upon satisfactory calibration, participants were immersed in a VR trial session. After completing the VR simulation with and without the mitigation techniques, participants were asked to complete an SSQ questionnaire [31] and then rest. We allowed participants to rest as long as they needed until they felt ready to continue the next session without any negative effects. The experiment was over when they completed applying all methods for minimizing the cybersickness, including the baseline condition (no mitigation). To validate our proposed approach and to compare it with the previously mentioned techniques, we designed five experimental conditions: 1) Initially, we ran the simulation as a baseline condition in which no techniques were applied, 2) Next, we applied the DFOV and DGB mitigation strategies to understand the tunneling and blur effects and 3) Then, as explained in Section 3.3, the RL-based adaptive cybersickness mitigation technique was applied with trained RL-policy, in which the policy recommended a suitable cybersickness mitigation technique either DFOV or DGB blur with appropriate intensity level, 4) After that, as explained in Sections 3.3 and 3.4, the user engaged with an LLM-based dialogue engine integrated with the trained RL policy, which provided explanations and allowed users to modify agent behavior, 5). Finally, in this condition, we used the RL policy fine-tuned through HIL adaptation using the data and user feedback collected from Condition 4. This personalized version of the RL agent was then evaluated with the same participants on a separate day to assess the effectiveness of personalized cybersickness mitigation in the AdaptVR (HIL) setting, as discussed in Sections 3.3, 3.4, and 3.5. It is worth mentioning that all the conditions were run in counterbalanced order. Moreover, each of the conditions lasted for 10 minutes. Figure 8 illustrates the first-person view of our experiment, where a participant is immersed in the AdaptVR framework and experiences RL-based DFOV mitigation. As shown, the mitigation effectively reduced discomfort without severely impacting the user's overall experience, as confirmed in post-experiment feedback. Furthermore, Figure 7(a), Figure 7(b), and Figure 7(c) present examples of the maze simulation outputs under three conditions: no mitigation, DFOV, and DGB. However, participants could stop at any point for any reason (e.g., if felt uncomfortable) during the experiment.

Post-task Interview Phase: In the post-task interview phase, we conducted an in-depth, qualitative interview with participants. First, they completed the post-SSQ. Afterward, they responded to a set of follow-up statements, such as: *"The AdaptVR framework helped reduce cybersickness and allowed me to stay engaged in the VR environment", etc.* We then asked participants about their experiences during the tasks, beginning with open-ended questions to avoid priming. Each interview lasted approximately 1–2 minutes and focused on participants' reflections on the study, particularly their experience with AdaptVR compared to conventional (without RL) mitigation methods. Before leaving, participants were reminded they could contact us if they experienced any severe discomfort or negative effects from the study. So far, no concerns have been reported from any participants.

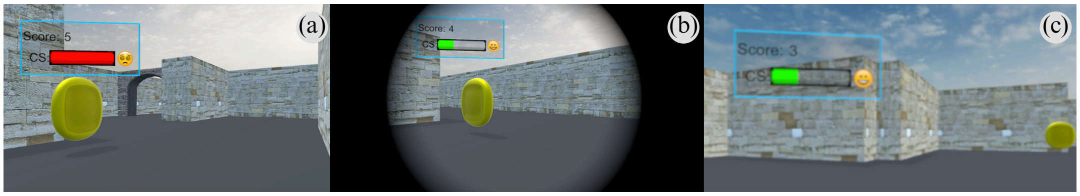

Figure 7: The user's perspective within the maze simulation when mitigation techniques are applied: (a) no mitigation, (b) DFOV, and (c) DGB.

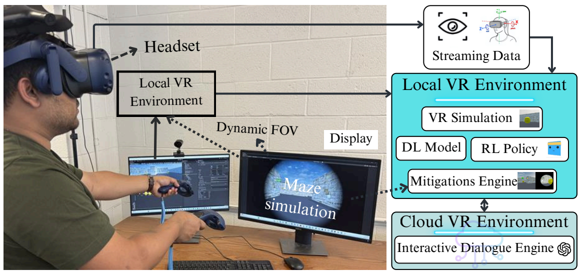

Figure 8: Participant immersed in the proposed AdaptVR framework.

6.2 Cybersickness Mitigation User Study Evaluation

This section compares the effects of RL-based cybersickness mitigation methods with other static/fixed mitigation techniques. We begin by analyzing our experimental data to examine users' responses to cybersickness when the RL-based approach was applied, and then we analyzed the post-task user study experimental data after encountering cybersickness in the VR experience. The details are described below.

Experiment phase: The mean, SD, and 95% CI of each condition's measured FMS score are shown in Table 2. The results show that the FMS score is lowest under the AdaptVR (HIL) condition, ranging from 1.17 to 2.37 within the 95% CI. Notably, both RL (1.38–2.74) and AdaptVR (1.30–2.56) yield significantly lower FMS scores compared to static methods. To test the significance of these mean differences, we first applied a Shapiro-Wilk test, which indicated that the FMS scores sampled every 30 seconds were not normally distributed. Therefore, we conduct a non-parametric Friedman test on these kinds of data instead of ANOVA. Note, we set our significance level $Alpha(\alpha) = 0.05$. The Friedman test indicates that there are significant differences in cybersickness between the conditions: the baseline, DFOV, DGB, RL, AdaptVR, and AdaptVR (HIL) (χ^2 = 40.45, p = 0.00001). For post hoc comparisons, we conduct Wilcoxon signed-rank tests with Bonferroni correction for each pair, as shown in Table 4. We observe that all mitigation techniques show a mitigation of cybersickness vs. no mitigation. Moreover, we find significant differences between each of the static mitigations vs. both AdaptVR and AdaptVR (HIL) - **DFOV vs. AdaptVR (**$Z = 3.92$**,** $W = 289$**,** $p = 0.0015$**)**, **DFOV vs. AdaptVR (HIL) (**$Z = 4.99$**,** $W = 202$**,** $p = 0.0010$**)**, **DGB vs. AdaptVR (**$Z = 3.31$**,** $W = 252$**,** $p = 0.0012$**)**, and **DGB vs. AdaptVR (HIL) (**$Z = 4.42$**,** $W = 159$**,** $p = 0.0003$**)**. However, we do not find any significant difference when comparing each DFOV with DGB mitigation (p-values > 0.05). Figure 9 shows the average FMS score of the different applied mitigation techniques in cybersickness mitigation for different users. The box plot shows that AdaptVR effectively reduces users' FMS scores, supporting **H1**.

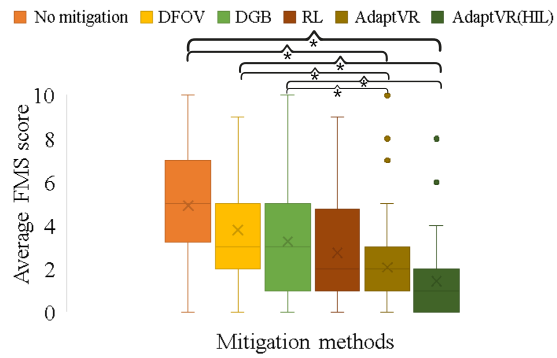

Figure 9: Average FMS scores across mitigation techniques (N=37).

Table 2: The mean, standard deviation (SD), and 95% CI of the FMS score for each condition (n=37).

Conditions	Mean (SD)	95% CI
No mitigation	4.60 (2.30)	[3.83, 5.37]
DFOV	3.80 (2.15)	[3.08, 4.52]
DGB	3.65 (2.20)	[2.92, 4.38]
RL	2.06 (2.05)	[1.38, 2.74]
AdaptVR	1.93 (1.90)	[1.30, 2.56]
AdaptVR (HIL)	1.77 (1.80)	[1.17, 2.37]

Furthermore, AdaptVR (HIL) achieves even greater effectiveness by personalizing mitigation to individual users, outperforming all other methods and providing strong support for **H3**.

Furthermore, to show the effectiveness of our proposed method, we calculate the effect size for each mitigation pair, as shown in Table 4. We observe medium effect sizes for **No Mitigation vs. (RL / AdaptVR)** and **Static vs. (RL / AdaptVR)**, and large effect sizes for **No Mitigation vs. AdaptVR (HIL)** and **Static vs. AdaptVR (HIL)**, indicating statistically significant and practically meaningful improvements supporting **H1** and **H3**. To evaluate different mitigation techniques for cybersickness, we also compute the mean FMS mitigation (ΔFMS), defined as the difference between the baseline (no mitigation) and each mitigation condition, following prior work [25, 39]. Table 3 shows the percentage mitigation achieved by each method. Static techniques DFOV and DGB achieve moderate mitigation levels of 55.7% and 61%, respectively. RL-based mitigation improves this to 72.8%, while AdaptVR further increases it to 75.5%. The highest mitigation is achieved with AdaptVR (HIL), reaching 83.9%, demonstrating the effectiveness of user-personalized adaptation. These findings further validate **H1** and **H3**, demonstrating that while adaptive mitigation is more effective than static approaches, incorporating user-specific preferences enhances its impact even further.

Table 3: Comparison among different mitigation techniques for reducing cybersickness

Mitigation techniques	ΔFMS
DFOV	55.7%
DGB	61.0%
RL	72.8%
AdaptVR	75.5%
AdaptVR (HIL)	83.9%

Table 4: Wilcoxon signed-rank tests with effect size for pairwise comparisons between mitigation conditions.

Condition Pair	Z-value	p-value	W-value	Effect size
No mitigation vs. DFOV	3.10	0.0023	422	0.35
No mitigation vs. DGB	2.50	0.0029	370	0.31
No mitigation vs. RL	4.13	0.0017	308	0.55
No mitigation vs. AdaptVR	5.13	**0.0010**	163	0.62
No mitigation vs. AdaptVR (HIL)	6.01	**0.00007**	102	0.85
DFOV vs. DGB	1.38	0.366	582	0.19
DFOV vs. RL	3.07	0.0022	486	0.60
DFOV vs. AdaptVR	3.92	**0.0015**	289	0.65
DFOV vs. AdaptVR (HIL)	4.99	**0.0010**	202	0.87
DGB vs. RL	3.20	0.0019	526	0.55
DGB vs. AdaptVR	3.31	**0.0012**	252	0.67
DGB vs. AdaptVR (HIL)	4.42	**0.0003**	159	0.79
RL vs. AdaptVR	1.95	0.133	457	0.12
RL vs. AdaptVR (HIL)	3.45	0.265	268	0.25
AdaptVR vs. AdaptVR (HIL)	2.33	0.087	151	0.10

Post-task interview Phase: We also provide additional user study results demonstrating how well the proposed AdaptVR framework provides comfort and effective cybersickness mitigation compared to the conventional methods. For this, during the user study, we asked users to rate their experiences with the AdaptVR framework on a 5-point Likert scale, compared to conventional frameworks (without RL-based mitigation) in which the applied mitigation methods are static, following prior work [37, 39]. More specifically, we asked users how much they agree with the following statements: **(S1)** *The AdaptVR framework helped reduce cybersickness and allowed me to stay engaged in the VR environment*, **(S2)** *The mitigation trigger made by AdaptVR felt reliable and appropriate based on my VR experience*, **(S3)** *Using the AdaptVR mitigation methods improved my overall VR experience compared to sessions without adaptive mitigations*, **(S4)** *I felt more comfortable using the AdaptVR mitigation techniques than static mitigations*, and **(S5)** *I would recommend the AdaptVR framework to others for its personalized and adaptive approach to cybersickness mitigation over conventional static methods.* We select these five specific statements to measure the proposed AdaptVR framework performance because they collectively capture the essential aspects of user experience and system effectiveness, which are critical for evaluating a new technology aimed at reducing cybersickness and enhancing VR engagement, following previous work [37, 39]. To evaluate the rated users' experiences, we compute the mean and standard deviation of the Likert score for these queries for each user. Note that we compute this with the AdaptVR framework and the conventional approach. The user feedback results are shown in Figure 10. We observe that approximately 90% of participants agreed with all statements to use our proposed AdaptVR framework for cybersickness mitigation and engaged in the VR simulation. At the same time, most users agree not to use the conventional approach due to ineffective UIX during immersion. Thus, these findings support **H2**, demonstrating

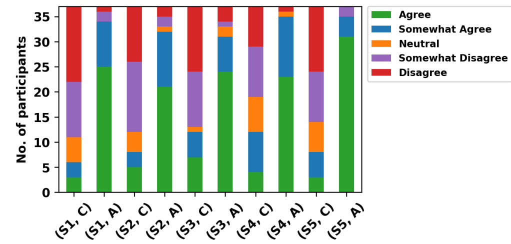

Figure 10: User feedback in Likert scale between the AdaptVR framework denoted as (A), and without the AdaptVR framework denoted as the conventional (C) technique, in which the mitigation technique is static.

that AdaptVR provides a better UIX in the virtual environment that enhances the overall VR experience.

SSQ results for all study conditions: In examining the SSQ scores, we compare the results from the pre-study and the post-study conditions for all participants and find a significant difference. For this, we first perform a Shapiro-Wilk test on these data, indicating that we cannot assume normality. Therefore, we conduct the Wilcoxon signed-rank test to quantify the difference between the pre and post-session SSQ subscores and the total scores. All subscores except Disorientation showed a significant post-session increase: Nausea($Z = -3.46$, $W = 88$, $p = 0.019$), Oculomotor($Z = -2.95$, $W = 59$, $p = 0.003$), Disorientation($Z = -1.65$, $W = 115$, $p = 0.078$), and TotalScore ($Z = -2.98$, $W = 60$, $p = 0.00015$).

7 Discussion

This section discusses the results of the AdaptVR framework. Both technical evaluations and user studies demonstrate that RL-based adaptive mitigation can significantly reduce cybersickness without compromising the UIX, which supports **H1**. Our results show that the proposed AdaptVR achieves a mean FMS mitigation of 75.5%. With HIL fine-tuning, mitigation improves further to 83.9%, highlighting the effectiveness of user-personalized adaptation and providing strong support for **H3** (see Table 3). Furthermore, the LLM-based interactive dialogue engine enables VR users to understand the causes of cybersickness and naturally express their individual preferences. This interactive capability led to higher user satisfaction, as approximately 90% of participants reported preferring AdaptVR over conventional static methods (see Figure 10).

Since no previous work has directly applied RL to mitigate cybersickness, our results are not directly comparable with prior studies [12, 19, 21, 25, 30, 34, 39, 69]. We therefore compared user comfort with AdaptVR against both static and dynamic mitigation techniques. Prior approaches to dynamic cybersickness mitigation have often relied on static methods with predefined presets [19, 26], which may hamper user immersive experience. For example, Kundu et al. [39] proposed an XAI-guided framework that maps important features to specific mitigation techniques. However, it also relies on predefined rules and lacks adaptive control over mitigation strength (e.g., DFOV aperture size or DGB intensity). As a result, it may also hamper the user immersive experience, increase cybersickness severity, and break immersion. In contrast, our RL-based adaptive mitigation approach learns to select the appropriate mitigation from the current VR state and adaptively adjusts its strength, rather than relying on static presets. For instance, while experiencing the static DFOV technique, participant **P7** reported: *"The fixed aperture felt restrictive and distracting. During turns and head rotations, I sometimes couldn't even see objects at the edge of my view."* In contrast, the same participant noted greater comfort with the RL-based DFOV condition, explaining: *"It felt much smoother because the aperture adapted naturally with my turns and rotations. I also liked that I could set my own preferences, which made the experience feel better."* On the following day, after experiencing the HIL fine-tuned version of AdaptVR, the same participant **P7** expressed even stronger satisfaction: *"Everything adjusted automatically to my needs, giving me the most natural and immersive experience overall."* Likewise, while experiencing the RL-based DGB technique, participant **P32** noted: *"I felt very comfortable because the system adjusted the blur strength based on my motion and movement duration."* In contrast, the same participant reported increased cybersickness with the static Gaussian blur condition, explaining: *"It felt like the system was trying to adjust the blur, but only within a fixed preset. This actually made things worse, leading to greater discomfort and reduced immersion. My visual clarity was also hampered unnaturally."* On the following day, after experiencing the HIL fine-tuned version of AdaptVR, participant **P32** expressed strong enthusiasm: *"I felt very comfortable as the system continuously provided mitigation with accurate intensity, aligning with my preferences without requiring instructions."* These results support both **H2** and **H3**, in which participants had a better UIX and personalized mitigation that enhanced the overall VR experience when using RL-based adaptive mitigation techniques. While the results for AdaptVR were largely positive, we also received a few critical comments that suggest areas for improvement. One participant noted that delays in responses from the LLM-based dialogue engine occasionally disrupted immersion. At the same time, another mentioned that although HIL adaptation improved comfort, the personalized adjustments did not always fully align with his preferences, leading to occasional mismatches in mitigation strength. Overall, AdaptVR effectively balances personalization, immersion, and comfort across users, demonstrating strong adaptability to diverse user profiles.

8 Limitation and future works

The AdaptVR framework has few limitations. First, our user study involved only a single population (users experienced only in a VR maze simulation) with imbalanced genders and did not consider other individual factors such as age, tasks, etc. This is important since cybersickness might affect people based on their unique characteristics, tasks, and VR environment. Different VR content and scenarios can have a direct influence on cybersickness. Hence, it is worth examining the effects of AdaptVR in other environments (e.g., games, interactive tasks, etc.) with longer exposure times, gender-balanced groups, and age-diverse participants. Second, the cybersickness detector used for the RL reward signal is primarily trained on maze data, and the RL policy is also developed in a VR maze simulation. Although domain randomization is applied across varied procedural contexts, the framework has not yet been evaluated in scenarios beyond this training environment. In the future, we plan to extend training to multiple environments, evaluate performance on unseen scenarios, and assess AdaptVR in more diverse contexts. Third, although our study focuses on evaluating the efficacy of AdaptVR in mitigating cybersickness during VR immersion, adding a presence questionnaire to capture users' sense of presence and engagement could have further improved the assessment from a user experience perspective. Finally, this work is limited to only two cybersickness mitigation techniques (DFOV and DGB). In the future, we plan to expand the mitigation library and explore continuous improvement mechanisms that leverage user feedback more effectively to make personalization safer and enhance long-term effectiveness.

9 Conclusion

This work proposed AdaptVR, an adaptive cybersickness mitigation framework that employs an RL agent to trigger mitigation actions adaptively, integrates an LLM-based dialogue engine to explain agent decisions and cybersickness reasons, and incorporates user preferences for cybersickness mitigation. Specifically, we developed an RL agent that adaptively adjusts the DFOV aperture and DGB intensity based on streaming eye-tracking, head-tracking, and scene context. This also allows users to understand and refine these mitigations through natural language interactions. To validate the framework, we implemented AdaptVR on the HTC Vive Pro HMD, released it as an open-source implementation, and conducted a user study comparing its performance with conventional static mitigation techniques. Results showed that AdaptVR significantly reduces cybersickness with minimal impact on UIX. Moreover, nearly 90% of participants found it intuitive and effective, particularly appreciating the ability to interact with agent responses via the LLM-based dialogue engine. These outcomes demonstrate that RL-based adaptive mitigation offers a promising direction for advancing cybersickness reduction on consumer-grade VR headsets.

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